

Set Theoretic Estimation and Language: Unification, Underspecification, and the Hyperframe Lattice

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Abstract

This note makes an intellectual case in three movements. First, a historical claim: set theoretic estimation (STE) in the sense of Combettes [4] is the estimation-theoretic framing of what two mature traditions in computational linguistics—unification grammar and underspecification semantics—already do operationally, and neither tradition cites Combettes; the bridge appears to be unclaimed. Second, a structural claim, now machine-checked in the Lean module `Ste.HyperFrame`: the STE satisfaction relation is a formal context in the sense of Formal Concept Analysis [7], partial feasibility sets are exactly its lower polars, and hence the closed feasible sets are the extents of a complete concept lattice—the *hyperframe lattice*. This yields, as corollaries, a disproof of naive oscillation (pure constraint accumulation is monotone) and a rough-set [8] indiscernibility partition that explains why Carlson’s AEP work partition [2] is STE-invariant. Third, an outlook: feasible sets glue over the constraint poset in a presheaf-like way, and a sheaf-theoretic representation is the natural next target; that direction is explicitly conjectural. Throughout we distinguish sharply between what is proven in the CI-verified Lean library (cited by identifier) and what is argued or conjectured.

1 The thesis: an unclaimed bridge

Combettes’s foundational formulation of set theoretic estimation [4] is austere: a solution space Ξ , an index set I of items of information, and for each $i \in I$ a *property set* $S_i \subseteq \Xi$ of solutions consistent with that item. The estimate is not a point but the *feasibility set* $\bigcap_{i \in I} S_i$; acquiring information shrinks it; a point estimate is warranted only in the *ideal* case where the intersection is a singleton.

Two traditions in computational linguistics rediscovered this schema operationally, in different guises, without the estimation-theoretic vocabulary and—as far as we have been able to determine—without citing Combettes or the set-membership estimation literature at all.

Unification grammar. In unification-based approaches to grammar [3, 9], partial information about a linguistic object is a *feature structure*, and combining two pieces of information is *unification*: the greatest lower bound in the subsumption order, when it exists. Each feature structure F denotes the set $\llbracket F \rrbracket$ of fully specified objects it subsumes, and the defining property of unification is precisely constraint intersection:

$$\llbracket F \sqcap G \rrbracket = \llbracket F \rrbracket \cap \llbracket G \rrbracket.$$

Unification failure is the empty intersection—in STE terms, an infeasible (over-constrained) problem. A parse succeeds ideally when the accumulated constraints cut the candidate space to a singleton. This is STE’s “intersect the property sets” scheme executed by a grammar, with subsumption playing the role of informativeness and $\sqcap$ playing the role of adding an item of information.

Underspecification semantics. In underspecification semantics the denotation of an ambiguous sentence is deliberately a *set of readings*, represented compactly and narrowed by constraints. Minimal Recursion Semantics [5] represents scope ambiguity by a flat bag of elementary predications plus handle constraints; each way of solving the constraints is one reading, so an MRS denotes the set of its solutions. Hole Semantics [1] and the Constraint Language for Lambda Structures [6] make the logic explicit: an underspecified representation is a constraint set, its models are the readings, and adding a dominance constraint monotonically prunes the model set. Again this is exactly the STE picture: readings are the solution space, each constraint is a property set, and the current semantic value of the discourse is the feasibility set.

The claim. The claim is not that these fields need STE to do what they already do. It is that STE is the common estimation-theoretic *frame*: both traditions perform set-valued estimation by constraint intersection, with monotone refinement, over-constraint (failure) and under-constraint (residual ambiguity) as the two failure modes, and the ideal case as a singleton feasibility set. Because neither literature cites Combettes, the identification—and with it the transfer of STE’s structural results (below) to grammar and semantics—appears to be an unclaimed bridge. The companion note on Constraint Grammar in this repository makes one instance of this identification fully formal; the present note supplies the lattice-theoretic superstructure.

2 Machine-checked: the hyperframe lattice

All results in this section are proven in Lean 4 over Mathlib [10] in the module `Ste.HyperFrame`, which builds in this repository’s CI; we cite each by its Lean identifier. Fix $S : I \rightarrow \mathcal{P}(\Xi)$, a family of property sets.

Definition 1 (satisfaction context; `STE.sat`). *The relation $\text{sat } S \subseteq \Xi \times I$ holds at (a, i) iff $a \in S_i$.*

This is literally a *formal context* in the sense of Formal Concept Analysis [7]: objects are hypotheses, attributes are constraints, incidence is satisfaction. The partial feasibility set $\Phi_J = \bigcap_{i \in J} S_i$ (only the constraints in $J \subseteq I$ enforced) then has an exact FCA description:

Theorem 1 (the bridge; `STE.partialFeasibilitySet_eq_lowerPolar`). *For every $J \subseteq I$, the partial feasibility set Φ_J equals the lower polar (attribute derivation J') of J under $\text{sat } S$.*

Consequently the Galois connection of the context organizes STE. A *hyperframe* of S (`STE.HyperFrame`) is defined as a formal concept of $\text{sat } S$: a pair (extent, intent) with extent the set of hypotheses satisfying every constraint in the intent, and intent the set of constraints satisfied by every hypothesis in the extent. Mathlib supplies the complete-lattice structure on concepts, so the hyperframes of S form a complete lattice with no further work. The two directions of the correspondence are:

Theorem 2 (feasible sets are extents; `STE.isExtent_partialFeasibilitySet`). *Every partial feasibility set Φ_J is the extent of a hyperframe.*

Theorem 3 (extents are feasible sets; `STE.hyperFrame_extent_eq_feasibilitySet`). *Every hyperframe’s extent equals the feasibility set of its own intent: $\text{extent}(c) = \Phi_{\text{intent}(c)}$.*

So the lattice of hyperframes is exactly the lattice of constraint-set/feasible-set fixpoints: the “reachable” estimation states of the problem, closed under the Galois closure. Enforcement is compositional:

Theorem 4 (superposition; `STE.partialFeasibilitySet_union`). $\Phi_{J \cup K} = \Phi_J \cap \Phi_K$.

For the linguistic reading: the hyperframe lattice is the space of stable “analysis states” of a grammar or of an underspecified semantic representation. Unification moves down this lattice (meet); the maximally underspecified representation is its top; unification failure is falling to an empty extent. None of the linguistic claims in Section 1 are formalized—what is formalized is the STE-side lattice they would land in.

3 Oscillation: the honest statement

A recurring informal worry about iterated constraint processing is *oscillation*: as information accumulates, does the feasible set thrash—shrinking to empty, then growing again? The machine-checked answer localizes the phenomenon precisely.

Theorem 5 (no oscillation under accumulation; `STE.ncard_partialFeasibilitySet_antitone`). *Over a finite hypothesis space, if $J \subseteq K$ then $|\Phi_K| \leq |\Phi_J|$: accumulating constraints can only shrink the feasible cardinality.*

Theorem 6 (oscillation requires retraction; `STE.lt_ncard_imp_not_subset`). *If $|\Phi_J| < |\Phi_K|$ then $J \not\subseteq K$: any growth of the feasible set forces the drop (retraction) of some previously enforced constraint.*

So *pure accumulation is monotone and cannot oscillate*; this disproves the naive form of the worry. What remains true—and this is the honest refinement—is that oscillation is exactly a *dynamic* phenomenon of edit streams that include retraction: reanalysis in parsing, defeasible constraints, belief revision. The repository’s `Ste.DynamicFrame` development studies that non-monotone regime; the present result fences it off cleanly from static STE.

4 Rough sets and Carlson’s AEP partition

The context `sat S` also carries Pawlak’s rough-set structure [8]. Define two hypotheses to be *indiscernible* (`STE.Indisc`) when they lie in exactly the same property sets.

Theorem 7 (`STE.indisc_equivalence`). *Indiscernibility is an equivalence relation on Ξ .*

Theorem 8 (granularity; `STE.indisc_mem_iff`). *Indiscernible hypotheses are co-feasible for every enforced constraint set J : $a \in \Phi_J \iff b \in \Phi_J$.*

Theorem 9 (signature fibers; `STE.indisc_iff_upperPolar_singleton_eq`). *a and b are indiscernible iff they have the same object intent: $\text{upperPolar}(\text{sat } S) \{a\} = \text{upperPolar}(\text{sat } S) \{b\}$. The Pawlak partition of Ξ is thus exactly the pullback partition of the FCA object-intent map, tying Section 2 and this section together.*

The equivalence classes are the *resolution floor* of the estimation problem: no subfamily of the available constraints can separate two indiscernible hypotheses, so every lower/upper (Pawlak) approximation, and every feasible set, is a union of classes. This is the STE reading of Carlson’s use of the asymptotic equipartition property in his cryptanalytic dissertation [2]: the AEP typical-set partition used there to divide the work of key search is an indiscernibility partition for the induced property sets, and the granularity theorem is the formal reason the partition is STE-invariant — constraint processing respects the classes, so work can be partitioned along them once and for all. (The identification of Carlson’s specific partition with `STE.Indisc` for his context is an interpretive claim of this note; what is machine-checked is the general indiscernibility theory just cited.)

5 Outlook: representation via sheaves

The superposition law $\Phi_{J \cup K} = \Phi_J \cap \Phi_K$ and antitonicity say that $J \mapsto \Phi_J$ is a limit-preserving contravariant assignment on the poset of constraint sets: local feasibility data over a cover $\{J_k\}$ of $J = \bigcup_k J_k$ determines feasibility over J , and compatible local sections should glue. The repository already contains a first mechanized step in this direction, `Ste.DynamicFrame.PresheafGluing`, which proves a gluing law for feasibility data over document/constraint structure.

Conjecture (outlook). *The assignment $J \mapsto \Phi_J$ extends to a sheaf on a suitable site (the constraint poset with covers given by unions, or a Grothendieck topology on the document poset in the dynamic setting), and the hyperframe lattice of Section 2 is recovered as its lattice of closed subsheaf sections. A representation theorem of this form would subsume both the concept-lattice picture and the dynamic gluing law.*

We flag this explicitly as *not yet proven*: no sheaf condition, site structure, or representation theorem is currently formalized, and we deliberately do not pursue sheaf- or homotopy-theoretic formalization here. It is the open direction this note points at.

References

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